

2.7.10

AI25BTECH11038 – Tejas Uppala

Question:

Find the area of the triangle whose vertices are $(-1, 1)$, $(0, 5)$, and $(3, 2)$

Solution:

The area of the triangle formed by three point vectors $\mathbf{A}$, $\mathbf{B}$ and $\mathbf{C}$ is

$$\text{area}(ABC) = \frac{\|(\mathbf{B} - \mathbf{A}) \times (\mathbf{C} - \mathbf{A})\|}{2} \quad (1)$$

Here,

$$\mathbf{A} = \begin{pmatrix} -1 \\ 1 \end{pmatrix}, \mathbf{B} = \begin{pmatrix} 0 \\ 5 \end{pmatrix}, \mathbf{C} = \begin{pmatrix} 3 \\ 2 \end{pmatrix} \quad (2)$$

$$(\mathbf{B} - \mathbf{A}) = \begin{pmatrix} 1 \\ 4 \end{pmatrix}, (\mathbf{C} - \mathbf{A}) = \begin{pmatrix} 4 \\ 1 \end{pmatrix} \quad (3)$$

$$\|(\mathbf{B} - \mathbf{A}) \times (\mathbf{C} - \mathbf{A})\| = 17 \quad (4)$$

Therefore,

$$\text{area}(ABC) = \frac{17}{2} \quad (5)$$

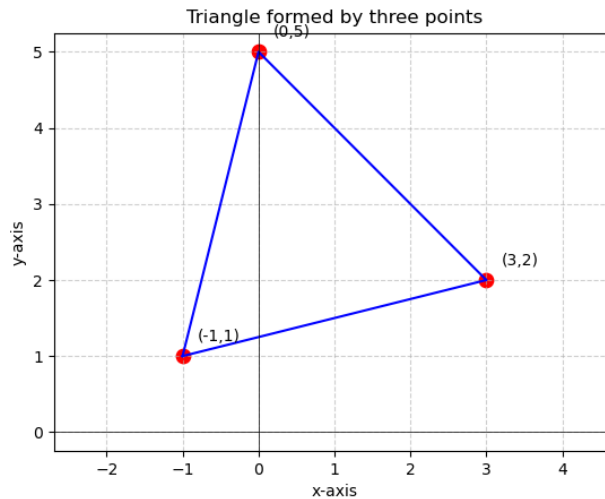


Fig. 1: The Plot